
MA 552: Number Theory

Course Code	MA 552
Course Name	Number Theory
L-T-P-C	3-0-0-3
Intended for	UG/PG
Distribution	Elective
One-time approval	9th BoA (Oct 2015); 9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Course Contents

- **Introduction & Divisibility Theory** [10 Lectures]
: Basics, Divisibility, Euclidean Algorithm, Primes and their Distribution, Prime Number Theorem (without proof). Congruences, Linear Congruences and Congruences with prime modulus, Some Diophantine Equations, The Chinese remainder theorem.
- **Number Theoretic Functions & Applications** [8 Lectures]
Arithmetic functions and the Mobius inversion formula, Greatest Integer Function, Sum of integer squares and Applications.
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- **Fermat's Theorem & Primitive Root** [8 Lectures]
Fermat's little theorem, Euler and Wilsons Theorems, Primitive Roots, Indices, Quadratic Reciprocity, Legendre Symbol, Gauss Theorem.
- **Applications to Cryptography & Special Topics** [10 Lectures]
Applications to Primality Testing, RSA & cryptography, Fibonacci Numbers, Numbers of Special Form, Continued Fractions and Rational Approximations.

Text Books

1. D. Burton, **Elementary Number Theory**, 7th edition, McGraw Hill, 2012.

References

1. Keneth Rosen, **Elementary Number Theory and its Applications**, 4th Edition, Addison-Wesley, 2000.
2. I. Niven, H.S. Zuckerman and Hugh L. Montgomery, **An Introduction to the Theory of Numbers**, 5th Edition, Wiley, 1991.